

Appendices

Appendix A

Some results from Calculus on $\mathbb{R}^n$

A.1 Results from Differentiation

Lemma A.1.1. *There exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is positive on $(-1, 1)$, 0 elsewhere and is C^∞*

Proof. Consider

$$f := \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & \text{if } x \in (-1, 1) \\ 0 & \text{elsewhere} \end{cases}$$

This function is C^∞ and is positive on $(-1, 1)$ and 0 elsewhere.

□

Lemma A.1.2. *Let $a \in \mathbb{R}^n$, $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by*

$$g(x) = f([x^1 - a^1]/\varepsilon) f([x^2 - a^2]/\varepsilon) \cdots f([x^n - a^n]/\varepsilon)$$

g is then C^∞ , and has positive values only on $(a^1 - \varepsilon, a^1 + \varepsilon) \times (a^2 - \varepsilon, a^2 + \varepsilon) \cdots \times (a^n - \varepsilon, a^n + \varepsilon)$.

Theorem A.1.3. (*Inverse Function Theorem*) *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. Suppose that $\det Df(a) \neq 0$ for a given $a \in \mathbb{R}^n$. Then, there exists U and V open in $\mathbb{R}^n$ such that $a \in U$, $f(a) \in V$ and $f|_U : U \rightarrow V$ is a diffeomorphism.*

Theorem A.1.4. (Implicit Function Theorem) Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be continuously differentiable in an open set of (a, b) and let $f(a, b) = 0$. Let M be the $m \times m$ matrix

$$D_{n+j}f^i(a, b); 1 \leq i, j \leq m$$

and suppose $\det M \neq 0$. Then there exists open sets $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^m$ with $a \in A$, $b \in B$ such that for all $x \in A$ there is a unique $g(x) \in B$ so that $f(x, g(x)) = 0$, and the g so obtained is differentiable.

Proof. Define $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n \times \mathbb{R}^m$ given by $F(x, y) := (x, f(x, y))$. This function is clearly differentiable, and moreover,

$$DF = \begin{pmatrix} \frac{\partial x^1}{\partial x^1} & \frac{\partial x^1}{\partial x^2} & \cdots & \frac{\partial x^1}{\partial x^n} & \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^m} \\ \frac{\partial x^2}{\partial x^1} & \frac{\partial x^2}{\partial x^2} & \cdots & \frac{\partial x^2}{\partial x^n} & \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x^n}{\partial x^1} & \frac{\partial x^n}{\partial x^2} & \cdots & \frac{\partial x^n}{\partial x^n} & \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^m} \\ \frac{\partial f^1}{\partial x^1} & \frac{\partial f^1}{\partial x^2} & \cdots & \frac{\partial f^1}{\partial x^n} & \frac{\partial f^1}{\partial y^1} & \frac{\partial f^1}{\partial y^2} & \cdots & \frac{\partial f^1}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} & \frac{\partial f^m}{\partial x^2} & \cdots & \frac{\partial f^m}{\partial x^n} & \frac{\partial f^m}{\partial y^1} & \frac{\partial f^m}{\partial y^2} & \cdots & \frac{\partial f^m}{\partial y^m} \end{pmatrix}$$

And it is easy to see that $\det DF(a, b) = \det M \neq 0$. Therefore, from Inverse function theorem (with little modification), there exists an open set $A \times B$ of $\mathbb{R}^n \times \mathbb{R}^m$, and an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ such that $(a, b) \in A \times B$, $(a, 0) \in W$ and $F : A \times B \rightarrow W$ is a diffeomorphism. $F^{-1} : W \rightarrow A \times B$ is also a diffeomorphism. Moreover, note that $F^{-1}(p, q)$ is of the form $(p, k(p, q))$ for some differentiable function $k : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$. $\pi \circ F(x, y) = \pi \circ (x, f(x, y)) = f(x, y)$ so $\pi \circ F \equiv f$. Look at $f \circ (p, k(p, q)) = f \circ F^{-1}(p, q) = \pi(p, q) = q$ so $f \circ (p, k(p, 0)) = 0$. Our $g(p)$ is given by $k(p, 0)$, and it is differentiable as a composition of differentiable functions.

□

Theorem A.1.5. (A corollary of A.1.4) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ ($p \leq n$) be continuously differentiable in an open set containing $a \in \mathbb{R}^n$. If $f(a) = 0$ and the matrix $D_j(f^i(a))_{p \times n}$ has rank p , then there exists an open neighbourhood A of a and a diffeomorphism $h : A \rightarrow h(A) \subset \mathbb{R}^n$ such that $f \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$.

Proof. Consider $f : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^p$. Suppose that the matrix $Q = D_{n-p+i}(f^j); i, j \leq p$ be invertible ($\det Q \neq 0$). Let, as before, $F : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^{n-p} \times \mathbb{R}^p$ given by

$$F(p, q) = (p, f(p, q))$$

Since $\det DF \neq 0$, there exists, around $a = (a^1, a^2, \dots, a^{n-p}, a^{n-p+1}, \dots, a^n)$ a neighbourhood $A_{n-p} \times A_p \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ and a neighbourhood $W \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ containing $F(a) = (a_1, a_2 \dots a_{n-p}, 0)$ such that $F : A_{n-p} \times A_p \rightarrow W$ is invertible and whose inverse is continuously differentiable. Note that, the inverse of F ought to be of the form $F^{-1}(x_1, x_2, \dots, x_{n-p}, y_{n-p+1}, \dots, y_n) = (\vec{x}, b(\vec{x}, \vec{y}))$ (let $\vec{x} = (x_1, x_2, \dots, x_{n-p}) \in \mathbb{R}^{n-p}$ and $\vec{y} = (y_{n-p+1}, y_{n-p+2}, \dots, y_n) \in \mathbb{R}^p$) where it is understood that $b : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is continuously differentiable. Also note that $\pi_2 \circ F \equiv f$. We have that

$$f \circ F^{-1}(x_1, x_2 \dots x_{n-p}, x_{n-p+1}, \dots, x_n) =$$

$$\pi_2(x_1, x_2, \dots, x_{n-p}, x_{n-p+1}, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$$

Case 2: If the indices where the matrix $\{(\partial_i f^j(a)); 1 \leq j \leq p, i = i_1, i_2, \dots, i_p\}$ is not the last p indices, we write the linear transformation $\text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that takes these p indices to the last p indices. This is a linear transformation that is invertible since it takes basis elements to basis elements. If we look then, at $f \circ \text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^p$, we see that this is a function of the form of Case 1, so we have a h such that $f \circ \text{Swap} \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$ which means our desired function is $\text{Swap} \circ h$.

□

A.2 Results from Integration

A.2.1 The Riemann Integral

Definition A.2.1. (Partition) For closed interval $[a, b]$ in $\mathbb{R}$, a partition is simply a collection of sub-intervals $[a_i, a_{i+1}]$ so that $a_0 = a < a_1 < a_2 < \dots < a_k = b$.

For a closed rectangle R in $\mathbb{R}^n$, a partition P , usually written along with its subpartitions (as in, $P \equiv (P_1, P_2, \dots, P_n)$), is a collection of subrectangles R_i (of a particular form) so that $R = \cup_i R_i$ and $R_i^\circ \cap R_j^\circ = \emptyset$. Explicitly, if

$$R = [a^1, b^1] \times [a^2, b^2] \times \dots \times [a^n, b^n]$$

,

$$P_i \equiv [a^i = a_0^i < a_1^i < a_2^i \cdots a_{k_i}^i = b^i]$$

and I_j^i denotes the j -th subinterval of the i -th partition, the collection of all possible sub rectangles of R of the form $\{I_{j_1}^1 \times I_{j_2}^2 \times \cdots \times I_{j_n}^n : 1 \leq j_q \leq \text{number of intervals}(P_q)\}$ is a partition of R .

Definition A.2.2. (Upper and Lower sums) Let R be a rectangle in $\mathbb{R}^n$, $f : R \rightarrow \mathbb{R}$ be a bounded function and let P be a partition of R .

$$\mathcal{L}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\inf_{x \in S} f(x) \right) |S|$$

$$\mathcal{U}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\sup_{x \in S} f(x) \right) |S|$$

where $|S|$ is the volume of the subrectangle S . For convenience, we denote $\inf_{x \in S} f(x)$ by $m_S(f)$ and $\sup_{x \in S} f(x) = M_S(f)$.

Lemma A.2.3. For a bounded function $f : R \rightarrow \mathbb{R}$,

$$\mathcal{L}(P, f) \leq \mathcal{U}(P, f)$$

for a given partition P .

Proof. If S_i are subrectangles of S coming from P' , then $m_S(f)|S| = m_S(f)(|S_1| + |S_2| + \cdots + |S_i|) \leq \sum_{S_i \in S} m_{S_i}(f)|S_i|$. This proves the result for lower sums, and the proof for upper sums is similar.

□

Lemma A.2.4. If P' is a refinement of P (i.e. each subrectangle of P' is contained in some subrectangle of P) then

$$\mathcal{L}(P, f) \leq \mathcal{L}(P', f)$$

and

$$\mathcal{U}(P', f) \leq \mathcal{U}(P, f)$$

Lemma A.2.5. If P and P' are any two partitions and $f : R \rightarrow \mathbb{R}$ a bounded function, then $\mathcal{L}(P', f) \leq \mathcal{U}(P, f)$.

Proof. Let P'' be a partition of both P and P' (i.e. the subrectangles of P'' are contained in some subrectangle of P as well as some subrectangle of P'). Then by the previous lemma,

$$\mathcal{L}(P', f) \leq \mathcal{L}(P'', f) \leq \mathcal{U}(P'', f) \leq \mathcal{U}(P, f)$$

□

Now we are in a position to define the Riemann Integral.

Definition A.2.6. ($\int_R f$) Let $f : R \rightarrow \mathbb{R}$ be a bounded function from a rectangle R . We say f is *Riemann integrable* if

$$\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f)$$

where $\sup$ and $\inf$ are taken over all partitions P of R . In that case we say $\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f) := \int_R f$.

Lemma A.2.7. A bounded function $f : R \rightarrow \mathbb{R}^n$ is Riemann Integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \varepsilon$.

A.2.2 Measure Zero and Content Zero

Definition A.2.8. (Measure Zero) A set $A \subset \mathbb{R}^n$ is said to be *measure zero* if for every $\varepsilon > 0$, there exists a countable collection of closed rectangles R_i such that $A \subset \bigcup_i R_i$ and $\sum_i |R_i| < \varepsilon$.

Lemma A.2.9.

1. Subsets of Measure Zero sets are Measure Zero.
2. Countable union of Measure Zero sets is Measure Zero.

Proof. (1) Is trivial.

For (2), consider E_i a countable collection of measure zero sets. Let $\varepsilon > 0$. Consider $\varepsilon/2^i$. There exists a collection of closed rectangles $\{R_{ij} : j \in \mathbb{N}\}$ such that $E_i \subseteq \bigcup_j R_{ij}$ and $\sum_j |R_{ij}| < \varepsilon/2^i$. Look at the collection $\{R_{ij} : i, j \in \mathbb{N}\}$ as a whole. This is a countable collection of closed rectangles that cover $\bigcup_i E_i$, and note that $\sum_{i,j} |R_{ij}| < \varepsilon$.

□

Definition A.2.10. (Content Zero) A set $A \subset \mathbb{R}^n$ is said to have *content zero* if for every $\varepsilon > 0$, there exists a *finite* collection of closed rectangles R_i such that $A \subseteq \cup_{i=1}^n R_i$ and $\sum_{i=1}^n |R_i| < \varepsilon$.

Lemma A.2.11. *If A is compact and has measure zero, then it has content zero.*

A.2.3 Integrable Functions

Definition A.2.12. (Oscillation) Let $f : X \rightarrow \mathbb{R}$ be a bounded function, where X is a metric space. Define

$$M(f, a, \delta) := \sup_{x \in B_\delta(a)} f(x)$$

and

$$m(f, a, \delta) := \inf_{x \in B_\delta(a)} f(x)$$

Let

$$\text{osc}(f, a, \delta) := M(f, a, \delta) - m(f, a, \delta)$$

Note first, that $\text{osc}(f, a, \delta)$ is decreasing for decreasing δ . Now we define

$$\boxed{\text{osc}(f, a) := \lim_{\delta \rightarrow 0} \text{osc}(f, a, \delta)}$$

An alternative characterisation of oscillation is the following:

Lemma A.2.13. *Let f be as in definition A.2.12. The quantity $\sup_{x, y \in B_\delta(a)} |f(y) - f(x)|$ is decreasing for decreasing δ and*

$$\text{osc}(f, a) = \lim_{\delta \rightarrow 0} \left(\sup_{x, y \in B_\delta(a)} |f(x) - f(y)| \right) = \inf_{\delta > 0} \left(\sup_{x, y \in B_\delta(a)} |f(x) - f(y)| \right)$$

Lemma A.2.14. *$f : X \rightarrow \mathbb{R}$ (where X is a metric space) is continuous at a if and only if $\text{osc}(f, a) = 0$*

Proof. $\implies$) Let f be continuous at a . For any given $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in B_\delta(a)$ we have

$$|f(x) - f(a)| < \varepsilon/2 \text{ and } |f(y) - f(a)| < \varepsilon/2$$

Adding these we get

$$|f(x) - f(y)| < \varepsilon$$

for $x, y \in B_{\delta(\varepsilon)}(a)$. Therefore, $\sup_{x, y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$. Since for every $\varepsilon > 0$, there exists some $\delta(\varepsilon) > 0$ such that $\sup_{x, y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$, we have $\text{osc}(f, a) = 0$.

$\Leftarrow$) Let $\text{osc}(f, a) = \inf_{\delta > 0} (\sup_{x, y \in B_\delta(a)} |f(x) - f(y)|) = 0$. This means that for every $\varepsilon > 0$, there exists $\delta > 0$ such that $\sup_{x, y \in B_\delta(a)} (|f(x) - f(y)|) < \varepsilon$, which means that

$$|f(x) - f(a)| < \varepsilon$$

whenever $x \in B_\delta(a)$.

□

Lemma A.2.15. (Recharacterisation of oscillation in terms of rectangles in $\mathbb{R}^n$)

$$\text{osc}(f, a) = \inf_{\delta > 0} \left(\sup_{x \in R_\delta(a)} f(x) - \inf_{x \in R_\delta(a)} f(x) \right)$$

Where $R_\delta(a)$ is a cube centred at a with side length δ .

Proof. Note that

$$B_{\delta/\sqrt{n}}(a) \subseteq R_\delta(a) \subseteq B_{\delta\sqrt{n}}(a)$$

From here the proof is trivial.

□

Lemma A.2.16. Let A be a closed rectangle and $f : A \rightarrow \mathbb{R}$ be a bounded function such that $\text{osc}(f, a) < \varepsilon$ for every $a \in A$. Then there exists a partition P of A such that $\mathcal{U}(f, P) - \mathcal{L}(f, P) < \varepsilon \cdot |A|$

Proof. We are given a fixed $\varepsilon_0 > 0$. We have that $\text{osc}(f, a) < \varepsilon_0$ for all $a \in A$. $\inf_{\delta > 0} (\sup_{x \in R_\delta(a)} f(x) - \inf_{y \in R_\delta(a)} f(y)) < \varepsilon_0$ for every $a \in A$. This means that, for every $a \in A$, there exists $\delta_a > 0$ such that $\sup_{x \in R_{\delta_a}(a)} f(x) - \inf_{x \in R_{\delta_a}(a)} f(x) < \varepsilon_0$. Collect all such $R_{\delta_a}(a)$. This is a cover of A , for which a finite subcover $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$ exists. Consider the partition P of A

formed by extending the grids of all the cubes $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$. The final partition P 's subrectangles J_i have the property that

$$\sup_{x \in J_i} f(x) - \inf_{x \in J_i} f(x) < \varepsilon_0$$

Essentially,

$$M_S(f) - m_S(f) < \varepsilon_0$$

for S (a subrectangle) $\in P$. This gives us that $\mathcal{U}(f, P) - \mathcal{L}(f, P) = \sum_{S \in P} (M_S(f) - m_S(f))|S| < \varepsilon_0 \sum_{S \in P} |S| = \varepsilon_0 |A|$

□

Theorem A.2.17. *Let A be a closed rectangle and $f : A \rightarrow \mathbb{R}$ be a bounded function. Let $B := \{x \in A : f \text{ not continuous at } x\}$. Then f is integrable if and only if B is a set of measure 0.*

Proof. Note that $B := \{x \in A : \text{osc}(f, x) > 0\}$ which could be written as $\cup_{n \in \mathbb{N}} \{x \in A : \text{osc}(f, x) \geq 1/n\}$, where we note that $\{x \in A : \text{osc}(f, x) \geq t\}$ is closed for any $t > 0$ (If $\text{osc}(f, a) < \varepsilon$ then there would exist a ball around a such that $\text{osc}(f, x) < \varepsilon$ for x in said ball).

□

Appendix B

Exterior Forms

B.1 Definitions and Results

Definition B.1.1. (Exterior Forms) Let $k \geq 1$ and $n \in \mathbb{N}$. A k -form on $\mathbb{R}^n$ is a map

$$f : \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{k \text{ copies}} \rightarrow \mathbb{R}$$

such that

1. f is linear in each of its k coordinates.
2. For every $\sigma \in \text{sym}(k)$, we have $f(x_{\sigma(1)}, x_{\sigma(2)} \cdots x_{\sigma(k)}) = \text{sgn}(\sigma)f(x_1, x_2, \cdots x_k)$.

Definition B.1.2. ($\Lambda^k(\mathbb{R}^n)$) We denote the space of all k -forms by $\Lambda^k(\mathbb{R}^n)$. By convention, we set

$$\Lambda^0(\mathbb{R}^n) = \mathbb{R}$$